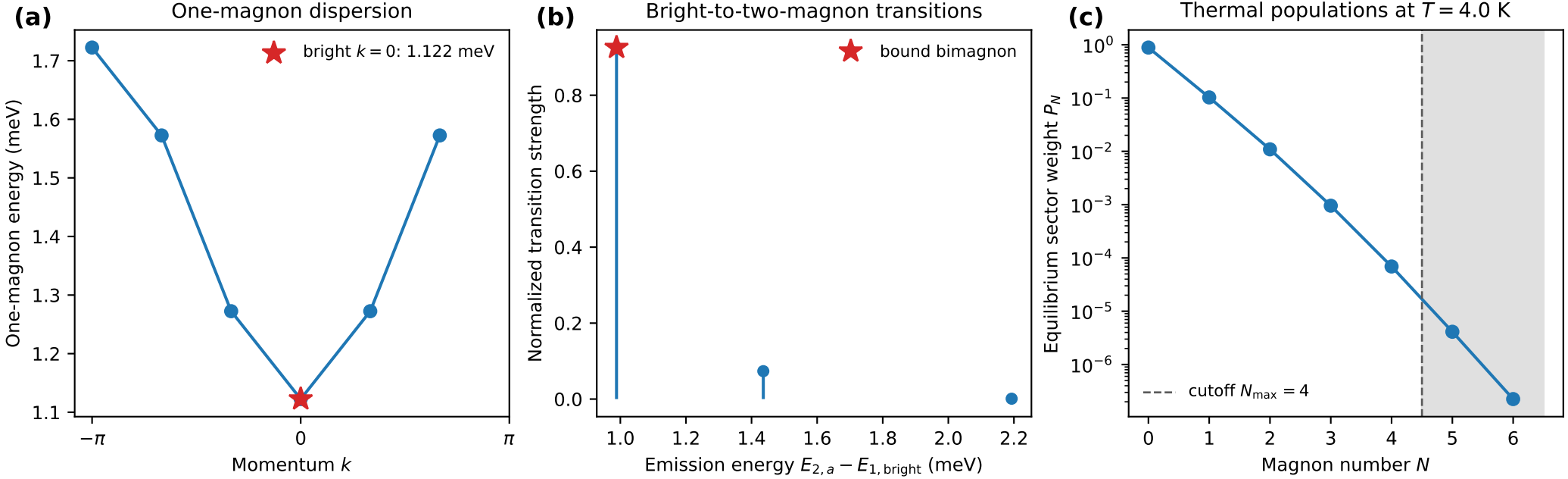
(a)
One-magnon dispersion
bright $k = 0$: 1.122 meV
One-magnon energy (meV)
1.7
1.6
1.5
1.4
1.3
1.2
1.1
$-\pi$
0
π
Momentum k
(b)
Bright-to-two-magnon transitions
bound bimagnon
Normalized transition strength
0.8
0.6
0.4
0.2
0.0
1.0
1.2
1.4
1.6
1.8
2.0
2.2
Emission energy $E_{2,a} - E_{1,\mathrm{bright}}$ (meV)
(c)
Thermal populations at $T = 4.0$ K
Equilibrium sector weight P_N
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
cutoff $N_{\max} = 4$
0
1
2
3
4
5
6
Magnon number N